

Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data

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Functional connectome has revealed remarkable potential in the diagnosis of neurological disorders, e.g. autism spectrum disorder. However, existing studies have primarily focused on a single connectivity pattern, such as full correlation, partial correlation, or causality. Such an approach fails in discovering the potential complementary topology information of FCNs at different connection patterns, resulting in lower diagnostic performance. Consequently, toward an accurate autism spectrum disorder diagnosis, a straightforward ambition is to combine the multiple connectivity patterns for the diagnosis of neurological disorders. To this end, we conduct functional magnetic resonance imaging data to construct multiple brain networks with different connectivity patterns and employ kernel combination techniques to fuse information from different brain connectivity patterns for autism diagnosis. To verify the effectiveness of our approach, we assess the performance of the proposed method on the Autism Brain Imaging Data Exchange dataset for diagnosing autism spectrum disorder. The experimental findings demonstrate that our method achieves precise autism spectrum disorder diagnosis with exceptional accuracy (91.30%), sensitivity (91.48%), and specificity (91.11%).

Key words: autism spectrum disorder; multi kernel learning; partial correlation; Pearson's correlation; Granger causality.

Introduction

Autism spectrum disorder (ASD) is a prevalent psychiatric condition characterized by neurological and developmental deficit (Bartlett et al. 2023; Easson et al. 2019). It has the potential to endure throughout an individual's lifespan and result in difficulties in engaging with others, expressing oneself, and exhibiting appropriate conduct (Alzheimer's Association 2022; Bamicha and Drigas 2022; Stevens et al. 2019). Currently, ASD poses a significant and escalating global health challenge, affecting an estimated 40–50 million individuals worldwide (Wen et al. 2022). Regrettably, the absence of a pathophysiological biomarker poses a significant challenge to accurately diagnosing ASD, as the current standard diagnosis heavily relies on symptomatology, thereby impeding treatment acceptance (Li et al. 2022). Indeed, early identification of ASD is imperative for optimizing therapeutic interventions and facilitating the recovery and enhancement of affected by this neurodevelopmental disorder.

With the rapid growth of neuroimaging techniques, such as resting-state functional magnetic resonance imaging (rs-fMRI), the integration of machine learning and medical imaging has become a new hotspot for exploring differences in brain activity between individuals with ASD and normal controls (NC; Chen and Zhao 2019; Shao et al. 2023; Wang et al. 2017). This approach facilitates the characterization of underlying causes, leading to improved diagnosis and treatment of neuro-diseases (Zhang et al. 2020). However, the challenge of identifying individuals with ASD

and NC based solely on fMRI remains due to the notorious issue of domain shift caused by random and asynchronous spontaneous brain activity between subjects and scanners (Li et al. 2017).

In contrast, with rs-fMRI, functional brain networks (FBN) have provided an alternative essential approach. In fact, FBN has been verified to be highly related to some neurological or psychological diseases such as ASD (Jiang et al. 2023; Li et al. 2017; Qiang et al. 2023; Unruh et al. 2023), MCI (Ding et al. 2023; Yang et al. 2023), and PD (Liu et al. 2023; Shi et al. 2023). Undoubtedly, such an approach can provide more stable measurements/biomarkers for the early diagnosis of ASD. However, most of the existing methods only consider a single connectivity pattern, such as full correlation, partial correlation, and causality (Li et al. 2021). Thus, in general, such approaches ignore the potential complementary topology information of FCNs at different connection patterns.

To address this issue, the present study focuses on integrating information from multiple connection patterns to enhance the accuracy of ASD diagnosis. In practice, we initially utilize rs-fMRI to construct functional brain networks based on the multiple connection patterns, including Pearson's Correlation (Li et al. 2017), Partial Correlation with Sparse Representation (Lee et al. 2011; Li et al. 2019; Ng et al. 2012; Yu et al. 2019), and Granger Causality (Goebel et al. 2003; Roebroeck et al. 2005). Specifically, inspired by multi-view learning in machine learning (Zhang et al. 2023; Zheng et al. 2023), we employ the Multi-Kernel Support Vector Machine (MKSVM) as a naïve attempt approach to combine information from various connection patterns (Li et al. 2023;

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Table 1. Demographic information of the subjects.

	ASD(N= 45)	NC(N= 47)	P-value
Gender (M/F)	36/9	36/11	0.2135
Age (year $\pm$ SD)	11.1 $\pm$ 2.3	11.0 $\pm$ 2.3	0.7773
FIQ (mean $\pm$ SD)	106.8 $\pm$ 17.4	13.3 $\pm$ 14.1	0.0510
ADI – R (mean $\pm$ SD)	32.2 $\pm$ 14.3*	–	–
ADOS (mean $\pm$ SD)	13.7 $\pm$ 5.0	–	–

ADI-R: Autism Diagnostic Interview-Revised; FIQ: Full Intelligence Quotient; ADOS: Autism Diagnostic Observation Schedule. *The P-value was obtained by chi-squared test.

Xu et al. 2020; Xu et al. 2021). Further, the proposed methods are validated through the conduction of an ASD classification task. In the end, the contribution of this work is highlighted as follows:

- (i) The proposed scheme, to the best of our knowledge, is the first effort in integrating multiple connection patterns to distinguish between ASD and NCs. Furthermore, empirical results unequivocally demonstrate that the proposed MCPC scheme exhibits remarkable superiority over single-pattern methods.
- (ii) The post-hoc analysis on the consensus connections generated from multiple connections is provided. Functional connectivity analyses are conducted to differentiate individuals with ASD from NCs and potentially elucidate the pathological mechanism of brain network underlying ASD.

Material and methods

Participants

In this study, we utilized a data set previously employed in another research project (Wee et al. 2016). The data set comprises resting-state fMRI (R-fMRI) information from 45 individuals diagnosed with ASD, aged between 7 and 15. All the data are publicly accessible through the ABIDE database (http://fcon_1000.projects.nitrc.org/indi/abide) (ABIDE I and II). Table 1 presents an overview of demographic details for all subjects involved. Diagnosis of ASD was based on autism criteria outlined in Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition, Text Revision (DSM-IV-TR), while psychopathology assessments were conducted via parent or participant interviews identify differential diagnosis and comorbidities-I disorders. Parent interviews relied on Schedule of Affective Schizophrenia for Children-Present and Lifetime Version (KSADS) for children under age 17.9 years old, whereas participants over age 18.0 years old underwent Structured Clinical Interview for DSM-IV Axis-I Disorders, Non-patient Edition (SCID-I) as well as Adult ADHD Clinical Diagnostic Scale (ACDS) interviews to determine psychopathology status. Exclusion from comorbid ADHD diagnoses required meeting all criteria except criterion E within DSM-IV-TR guidelines; inclusion as a neurotypical control necessitated exclusion based on current KSADS-PL, SCID-I/NP, and ACDS interview results.

fMRI data acquisition

The R-fMRI images were acquired using a standard echo-planar imaging sequence on a 3 T Siemens Allegra scanner commonly employed in clinical settings. During the 6-min R-fMRI scanning procedure, participants were instructed to relax while fixating their gaze on a white fixation cross positioned at the center of a black background. The imaging parameters included a flip angle of 90°, acquisition of 33 slices, and a TR/TE of 2000/15 ms with the collection of 180 volumes. The voxel thickness was set at 4.0 mm.

Neuroimaging data pre-processing

To preprocess the data, we utilized Statistical Parametric Mapping, which is widely recognized software for this purpose. Specifically, we discarded the initial ten R-fMRI to minimize any potential confounding effects. The remaining images underwent calibration through several steps: (i) transformation into MNI space with a resolution of $3 \times 3 \times 3 \text{ mm}^3$; (ii) regression analysis that accounted for nuisance signals such as ventricle global signals, and head motion using Friston's well-established model consisting of 24 parameters (Friston et al. 1996); (iii) band-pass filtering between frequencies ranging from 0.01 to 0.08 Hz; and, finally, (iv) signal de-trending was performed to remove any linear trends present in the data over time. Subsequently, we divided the pre-processed BOLD time series signals into regions of interest (ROIs).

Brain network construction

To construct brain networks using various connection patterns, we divided the pre-processed rs-fMRI time series signals into 90 ROIs based on the AAL atlas. Specifically, denoting $x_i \in R^t$ as the observed time course associated with i th brain regions, where t is the number of time nodes. Correspondingly, $\bar{x}_i \in R^t$ represents the mean value of the elements in x_i , $i = 1, 2, \dots, n$, and n is the number of ROIs. Thus, $x_i - \bar{x}_i$ is a centralized counterpart of x_i . After that, the networks were then constructed by the Partial correlation (Sparse Representation for sparse constraint), Pearson's Correlation, and Granger causality, whose definitions are given as below:

Pearson's Correlation

The FBN construction commonly employs Pearson's Correlation, which effectively captures the comprehensive correlation among ROI (Langer et al. 2012). More precisely, the mathematical representation of PC is defined as follows:

$$W_{ij} = \frac{(x_i - \bar{x}_i)^T (x_j - \bar{x}_j)}{\sqrt{(x_i - \bar{x}_i)^T (x_i - \bar{x}_i)} \sqrt{(x_j - \bar{x}_j)^T (x_j - \bar{x}_j)}}, \quad (1)$$

Partial Correlation

The Pearson's Correlation measures the full correlation cross ROIs (Langer et al. 2012; Lee et al. 2011; Li et al. 2017; Ng et al. 2012; Yu et al. 2019), which suffers from the issue of confounding effect. In contrast, the partial correlation based FBN estimation filtering out the confounding variant to avoid such an effect. However, the solvers of partial correlation are often ill-posed. Thus, a sparse regularization (SR) term is often conducted as a constraint to solve the partial correlation-based FBN:

$$W = \arg \min_W \sum_{i=1}^n \left\| x_i - \sum_{j \neq i} W_{ij} x_j \right\|^2 + \lambda \sum_{j \neq i} |W_{ij}|, \quad (2)$$

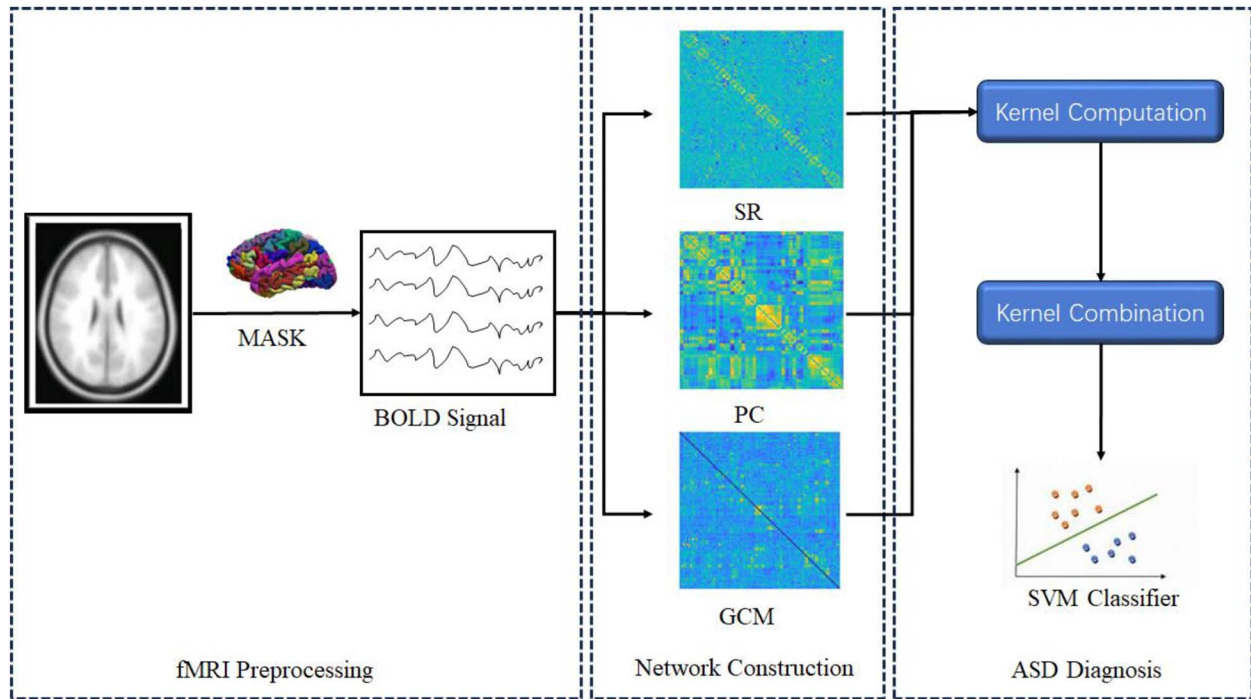


Fig. 1. The entire pipeline of multiple connection patterns for ASD diagnosis.

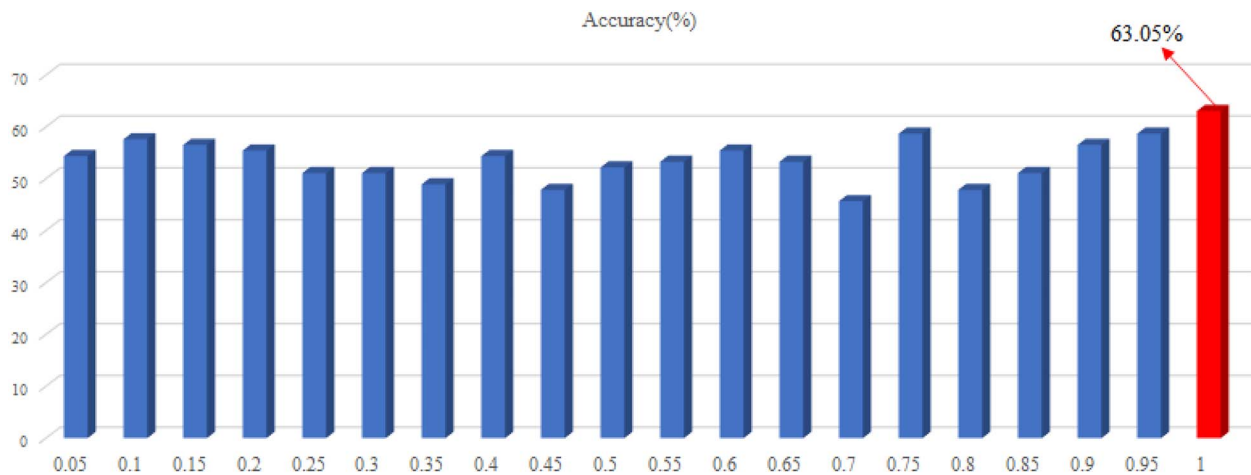


Fig. 2. The performance of SR for different hyperparameter λ .

where λ is the hyper-parameter to control the balance between two terms.

Granger Causality

Both PC and SR measure the correlation cross ROIs, which is undirected network. In contrast, Granger Causality measures the causality between ROIs (Goebel et al. 2003). In particular, Granger Causality measures the predictive ability of an earlier time series from one ROI to the future value of another lagging time series of another ROI. Thus, it provides a directed graph of brain network and thereby nonsymmetric. Specifically, given two-time series $x_i[n]$ and $x_j[n]$, the Granger causality-based FBN is defined as follows:

$$W_{ij} = G_{x_i \rightarrow x_j} = \ln \frac{|\sum (\zeta_i)|}{|\sum (\zeta_j)|}, \quad (3)$$

where ζ_i and ζ_j are the residuals of the restricted and unrestricted regression models (w/o $x_j[n]$), respectively, and $\sum(\cdot)$ indicates the variance.

Multi kernel classification for ASD diagnosis

The most straightforward approach to integrating information from multiple connectivity patterns is to directly concatenate all the data. However, in the case of high-dimensional curves and small sample sizes, this direct splicing method can lead to a curse of dimensionality disaster and pose a risk of performance degradation. To address this issue, we propose utilize the idea from the multi view learning. Specifically, we adopt an MKSVM approach, which leverages kernel fusion techniques to integrate the information form different connection pattern. The specific framework is illustrated in Fig. 1.

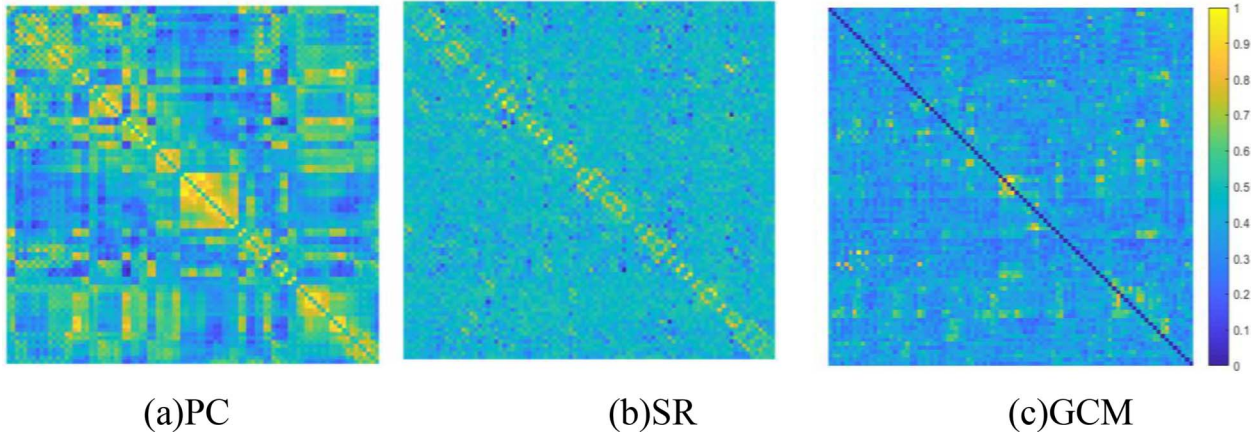


Fig. 3. The constructed brain networks obtained by the (a) PC, (b) SR, and (c) GCM methods.

Furthermore, the original problem encountered when applying the kernel fusion method in MK-SVM, as follows.

$$\begin{aligned}
 & \min_W \frac{1}{2} \sum_{m=1}^M \beta_m \|W^m\|^2 + C \sum_{i=1}^n \xi_i, \\
 & \text{s.t. } y_i \left(\sum_{m=1}^2 \beta_m (W^m)^T \varphi^m(x_i^m) + b \right) \geq 1 - \xi_i, \\
 & \xi_i \geq 0, \quad i = 1, 2, \dots, n
 \end{aligned} \quad (4)$$

where n is the number of the participants, $y_i \in \{1, -1\}$ represents the label (ASD vs NC) of the corresponding, i.e. i th participant. The number of the adopted connection patterns is denoted as M , where $M = 3$ in this paper corresponding to PC, SR, and GCM. W^m denotes the hyper-plane in the reproducing kernel Hilbert space as well as β_m represents the combination weight of the m th pattern. Formally, we also have the dual form of the MK-SVM:

$$\begin{aligned}
 & \max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{ij} \alpha_i \alpha_j y_i y_j \sum_{m=1}^M \beta_m k^m(x_i^m, x_j^m), \\
 & \text{s.t. } \sum_{i=1}^n \alpha_i y_i = 0, \\
 & 0 \leq \alpha_i \leq C, i = 1, 2, \dots, n.
 \end{aligned} \quad (5)$$

where $k^m(x_i^m, x_j^m) = \varphi^m(x_i^m)^T \varphi^m(x_j^m)$ represents the kernel matrix estimated by the m th pattern. After we trained/obtained the final MK-SVM classifier, the diagnostic result of new participant $\{x_i\}_{i=1}^M$ can be obtained as follows:

$$f(x_1, x_2, \dots, x_M) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i \sum_{m=1}^M \beta_m k^m(x_i^m, x^m) + b \right). \quad (6)$$

Results

Brain network visualization

The BrainNetClass tool (Zhou et al. 2019) was utilized to estimate the brain network based on PC and SR methods. It should be noted that the SR method involves a hyperparameter λ . Therefore, we employed leave-one-out cross-validation (LOOCV) to determine the optimal value for λ in order to construct the SR-based brain network, we determined the optimal value for λ through LOOCV,

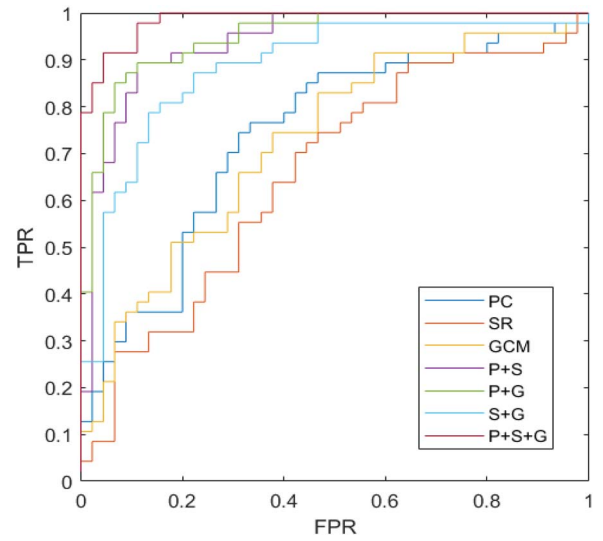


Fig. 4. The ROCs of different methods. False positive rate (FPR), true positive rate (TPR).

within a specific range in $[0.05 : 0.05 : 1]$. Empirically, we set $\lambda = 1$ for the final SR-based FBN construction, which resulted in an accuracy of 63.05%. Figure 2 presents the accuracies obtained with different values of λ using LOOCV. The dynamicBC toolbox was conducted for constructing the GCM-based brain network (Liao et al. 2014).

We depicted the estimated FBN adjacency matrices (to facilitate comparison among PC and SR methods, all weights are normalized within the range of $[-1, 1]$ for enhanced convenience) of PC, SR, and GCM methods in Fig. 3. The brain connections obtained by different brain network estimation methods exhibit distinct topologies in Fig. 3, as these methods capture diverse statistical information or relationships across ROIs.

Classification results

Due to the limited sample size, we employ the nested LOOCV strategy to assess the performance of MCI classification (Li et al. 2020). Specifically, an inner LOOCV is conducted to determine the optimal parameters, namely the hyperparameter C in SVM. The range of hyperparameter C is set as $\{2^{-5}, 2^{-4}, 2^{-3}, 2^{-2}, 2^{-1}, 1, 2^1, 2^2, 2^3, 2^4, 2^5\}$. Furthermore, accuracy, sensitivity, specificity, and AUC are utilized for evaluating

Table 2. Demographic information of the subjects.

Method	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC
PC(P)	70.65	76.59	64.44	0.7404
SR(S)	63.05	63.82	62.22	0.6553
GCM(G)	66.30	68.09	64.44	0.7258
P + S	85.86	82.97	88.89	0.9371
P + G	89.13	89.36	88.89	0.9779
S + G	81.52	76.59	86.67	0.8836
P + S + G	91.30	91.48	91.11	0.9853
Concat_PSG	86.95	88.89	85.10	0.9314

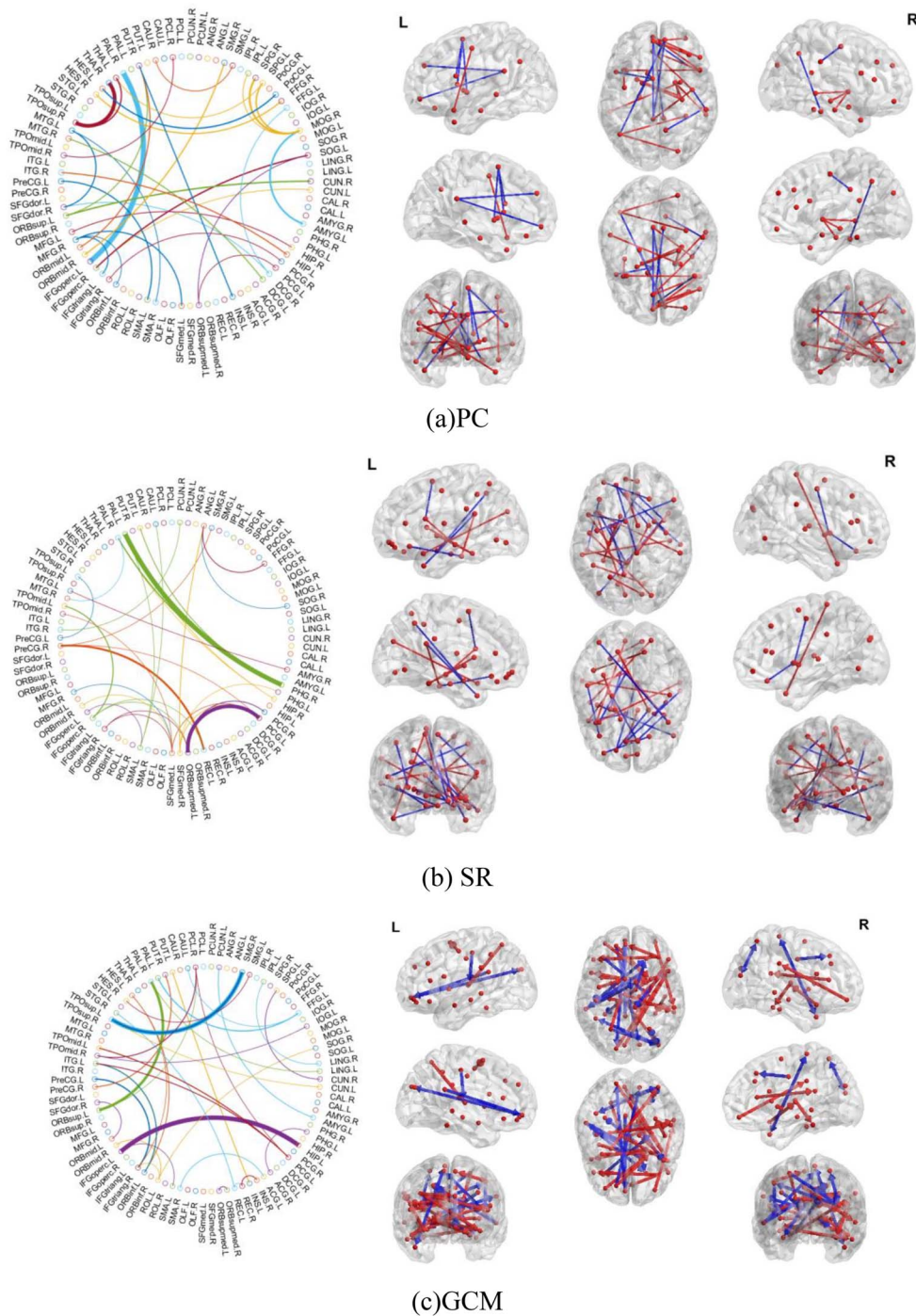


Fig. 5. The consensus connections on different connection patterns.

Table 3. In the context of PC method, this study identifies the top 10 consensus connection.

ROI	ROI	P-value	NC	ASD
PHG.R	PAL.L	3.99×10^{-6}	0.006487	0.224671
ORBsupmed.L	PCG.L	5.20×10^{-6}	0.573024	0.397562
PreCG.R	REC.L	1.09×10^{-5}	-0.18451	0.011231
PHG.R	PAL.R	2.14×10^{-5}	0.034155	0.23144
IFGoperc.R	TPOmid.L	2.94×10^{-5}	-0.20138	-0.03773
ORBsupmed.L	PCG.R	4.66×10^{-5}	0.472743	0.293983
FFG.R	ANG.L	5.06×10^{-5}	-0.32496	-0.18894
PHG.R	PUT.L	5.09×10^{-5}	0.048192	0.2362
SFGmed.R	ANG.L	5.35×10^{-5}	0.379337	0.202937
SFGmed.R	PCG.L	7.27×10^{-5}	0.36551	0.203193

Table 4. In the context of SR method, this study identifies the top 10 consensus connection.

ROI	ROI	P-value	NC	ASD
SMG.R	TPOsup.L	1.95×10^{-4}	-0.06422	0.013943
PAL.R	ORBsup.L	3.65×10^{-4}	0.078615	-0.02995
IFGoperc.L	HIP.L	4.98×10^{-4}	-0.04347	0.030749
ORBsup.L	PAL.R	1.01×10^{-3}	0.041331	-0.02103
PCG.L	TPOmid.L	1.07×10^{-3}	0.100951	-0.01128
ORBinf.L	ITG.L	1.15×10^{-3}	0.074447	0.000391
PUT.R	PreCG.R	1.64×10^{-3}	0.025011	-0.02895
ITG.L	PCG.L	1.75×10^{-3}	0.00344	0.037597
TPOmid.L	PCG.L	1.96×10^{-3}	0.056003	-0.0111
PCUN.L	AMYG.L	2.33×10^{-3}	0.022531	-0.01923
ORBmid.R	ROL.L	2.70×10^{-3}	0.008507	-0.08374

Table 5. In the context of GCM method, this study identifies the top 10 consensus connection.

ROI	ROI	P-value	NC	ASD
IFGoperc.L	PAL.R	3.71×10^{-4}	0.006611	0.021689
THA.R	TPOsup.R	5.33×10^{-4}	0.013601	0.038598
SPG.R	MOG.R	1.22×10^{-3}	0.020418	0.0092
MOG.R	AMYG.L	1.30×10^{-3}	0.007508	0.020672
CUN.R	ORBsup.L	1.70×10^{-3}	0.016715	0.00701
MOG.R	SPG.L	1.79×10^{-3}	0.021709	0.008306
IFGoperc.R	SOG.L	1.88×10^{-3}	0.007162	0.017853
HES.L	PoCG.L	2.05×10^{-3}	0.026529	0.014607
THA.L	TPOsup.R	2.07×10^{-3}	0.01372	0.032929

of different measurements. AUC is the Area under the ROC curve. The mathematical definitions of accuracy, sensitivity, and specificity are provided below.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}, \quad (7)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad (8)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (9)$$

where the variable TP (represents the count of correctly classified positive subjects in the task of identifying ASD. Similarly, TN (True Negative), FP (False Positive), and FN (False Negative) denote the respective counts for their corresponding subjects.

The classification results obtained from distinct connection patterns are displayed in Table 2, which were accomplished using a traditional linear kernel SVM classifier. Additionally, the

outcomes achieved by merging partial connection patterns (e.g. PC + SR, SR + GCM) are also included. Specifically, to validate the performance of the multi-kernel methods, we also provided the concat_PSG method, which simple concatenate the feature of different pattern to train a single kernel SVM. The ROC curve can be observed in Fig. 4.

Consensus connections

Considering the potential confounding effect on the final classification outcome, we employed a straightforward feature selection method, namely t-test. Since the selected features/connections may vary across different validation loops, we identified the consensus connection as the most discriminative one in each connection pattern (as depicted in Fig. 5). Specifically, there were 31 connections in the PC-based brain network, 26 connections in the SR-based network, and 35 connections in the GCM-based network. GCM method exhibits the maximum number of consensus connections.

Table 6. Hubs of the NC and ASD groups based on the PC method.

	AAL number	Corresponding brain region	Subnetwork
NC	29	INS.L	SN
	30	INS.R	SN
	81	STG.L	VAN
	17	ROL.L	AN
	23	SFGmed.L	DMN
	49	SOG.L	DAN
	18	ROL.R	AN
	50	SOG.R	DAN
	51	MOG.L	DMN
	47	LING.L	DMN
	24	SFGmed.R	DMN
	56	FFG.R	DMN
	46	CUN.R	VN
	52	MOG.R	VN
	48	LING.R	VN
	3	SFGdor.L	DMN
	25	ORBsupmed.L	DMN
ASD	29	INS.L	SN
	30	INS.R	SN
	81	STG.L	VAN
	49	SOG.L	DAN
	23	SFGmed.L	DMN
	74	PUT.R	Subcortical
	48	LING.R	VN
	50	SOG.R	VN
	24	SFGmed.R	DMN
	47	LING.L	VN
	82	STG.R	VAN
	17	ROL.L	CTC
	51	MOG.L	VN
	43	CAL.L	VN
	18	ROL.R	AN
	73	PUT.L	Subcortical
	46	CUN.R	VN
	52	MOG.R	VN

SH: sensory/somatomotor hand; VAN: ventral attention network; AN: auditory network; DMN: default mode network; DAN: dorsal attention network; VN: visual network.

More specifically, we have also included the connection weight of these consensus connections on various connection patterns in Tables 3–5. Referring to the P-value, it can be found that the group difference in consensus connections obtained by the PC method is more significant between NC and ASD patients. However, due to space constraints, we have only presented the top 10 most significant consensus connections for each connection pattern based on the p-value obtained from t-test analysis. For more detailed information, please refer to the supplementary files. In terms of altered patterns, we found that functional brain networks based on these three methods exhibited enhanced consensus connections between the cortex and subcortical nuclei in ASD patients. For instance, the consensus connections between the temporal lobe, frontal lobe, limbic system (parahippocampal gyrus, posterior cingulate gyrus) and subcortical nuclei (pallidum, putamen, amygdala) were mostly enhanced. However, the connections between the frontal lobe, limbic system, and parietal lobe in ASD patients were decreased. More details can be found in Figures S1–S6 and Tables S1–S3.

Distribution of hubs

Referring to previous studies (Peng et al. 2022), we identified the top 5% of brain regions with the largest weight as the hubs of

group-level brain networks. As shown in Tables 6–8, there was no significant difference in the distribution of hubs between the ASD and NC groups based on the SR method. Nevertheless, based on functional brain networks constructed by PC and GCM methods, we found that the distribution pattern of hubs was altered between ASD patients and NC. Specifically, the number of hubs distributed in the DMN significantly decreased, while the number of hubs distributed in the VN and subcortical increased.

Discussion

The classification performance of different network construction approach

Based on the findings presented in Table 2 and Fig. 4, it is evident that the performances that of the single-kernel SVM. These results strongly support the validity of our proposed MCPC approach. To assess the significant improvement in model performance, we conducted a Delong test (Delong et al. 1988) to compare differences between various AUCs. The results demonstrate that our MCPC methods outperform individual patterns such as PC, SR, and GCM with high confidence levels (P-values equal to 0.0081, 0.0116, and 0.00043, respectively, under a 95% confidence interval). This superior performance clearly indicates that our

Table 7. Hubs of the NC and ASD groups based on the SR method.

	AAL number	Corresponding brain region	Subnetwork
NC	41	AMYG.L	Subcortical
	25	ORBsupmed.L	DMN
	80	HES.R	AN
	42	AMYG.R	Subcortical
	21	OLF.L	Uncertain
	35	PCG.L	Memory retrieval
	79	HES.L	AN
	27	REC.L	DMN
	54	IOG.R	VN
	9	ORBmid.L	FTC
	36	PCG.R	DMN
	70	PCL.R	SH
	65	ANG.L	DMN
	10	ORBmid.R	FTC
ASD	41	AMYG.L	Subcortical
	9	ORBmid.L	FTC
	25	ORBsupmed.L	DMN
	80	HES.R	AN
	21	OLF.L	Uncertain
	42	AMYG.R	Subcortical
	27	REC.L	DMN
	35	PCG.L	Memory retrieval
	65	ANG.L	DMN
	10	ORBmid.R	FTC
	79	HES.L	AN
	36	PCG.R	DMN
	54	IOG.R	VN
	70	PCL.R	SH

DMN: default mode network; FTC: frontoparietal task control.

MCPC approach significantly enhances classification accuracy even when using only single-modal data. Furthermore, it is worth noting that despite relying solely on single-mode data, our so-derived approach still achieves remarkable improvements in ASD diagnosis accuracy. Additionally, for the three different brain network construction methods (i.e. PC, SR, GCM), our results suggest that the group difference in the consensus connections obtained by the PC method was more significant between NC and ASD patients, which might potentially account for the better classification performance of PC method.

The altered pattern of the prominent features

From Tables 3–5, we can easily observe that the consensus connection between ASD and NC from different connection patterns are significantly different, which confirms that the connection information from different pattern is different. Such results indicate that the combination of different connection patterns is necessary. It can be easily seen that in three connection patterns, the consensus connections from IFGoperc and PAL all exist. These three methods exhibited enhanced consensus connections between the cortex and subcortical nuclei in ASD patients. Specifically, we can find that in the consensus connectivity obtained by PC, the connections between PAL.L and PHG.R were significantly higher in ASD than in NC. The connectivity between PreCG.R and REC.L in ASD was also higher than that in NC. In the consensus connection obtained through SR, the difference between ASD and NC with IFGoperc.L and HIP.L was also a little larger than the difference between SMG.R and TPOsup.L. Further, in the results of the connectivity between PAL. R and IFGoperc.L, ASD was also higher than that in NC. It is worth noting that

according to Tables 3–5, the difference between ASD and NC in brain area connectivity is not all unilateral increase, and ASD with many brain regions connectivity is lower than the NC. For example, the connection between ORBinf.L and ITG.L in Table 4 is lower than that in NC, and the same is true of the connection between SPG.R and MOG.R in Table 5. The results are also consistent with some current studies (Guo et al. 2023; Ma et al. 2023). Meanwhile, we obtained the hubs of group-level brain networks, and project these brain regions into functional subnetworks. Our results indicated that most of hubs based on PC, SR and GCM were involved in the DMN, SN, VN, subcortical and attention network. In particular, compared with the NC group, the number of hub nodes distributed in the DMN was significantly reduced in ASD patients, while the number of hub nodes distributed in the VN and subcortical subnetwork was significantly increased. These sub-networks play an important role in learning and memory, emotional expression, attention, and behavior control (Liu et al. 2023; McLoughlin 2023; Sato and Uono 2019).

Conclusion

In this study, we aim to enhance the performance of ASD identification using single-mode data by generating multi-view information. Specifically, we leverage information derived from multiple brain connection patterns obtained through fMRI data analysis. To achieve this, we employ MKSVM as a naive approach for identifying ASD from NCs, effectively integrating the information from diverse brain connection patterns. The experimental results demonstrate that our strategy significantly improves diagnostic performance compared to using a single pattern alone.

Table 8. Hubs of the NC group based on the GCM method.

	AAL number	Corresponding brain region	Subnetwork
NC	78	THA.R	Subcortical
In degree	3	SFGdor.L	DMN
	77	THA.L	Subcortical
	70	PCL.R	SH
	72	CAU.R	Subcortical
	44	CAL.R	VN
	71	CAU.L	Subcortical
	30	INS.R	SN
	23	SFGmed.L	DMN
	81	STG.L	VAN
	67	PCUN.L	DMN
	57	PoCG.L	SH
	36	PCG.R	DMN
	31	ACG.L	DMN
	8	MFG.R	FTC
Out degree	77	THA.L	Subcortical
	78	THA.R	Subcortical
	32	ACG.R	DMN
	39	PHG.L	DMN
	71	CAU.L	Subcortical
	44	CAL.R	VN
	47	LING.L	DMN
	76	PAL.R	Subcortical
	83	TPOsup.L	CTC
	48	LING.R	VN
	70	PCL.R	SH
	69	PCL.L	SH
	72	CAU.R	Subcortical
	31	ACG.L	DMN
	88	TPOMid.R	DMN
ASD	78	THA.R	"Subcortical"
In degree	77	THA.L	"Subcortical"
	30	INS.R	SN
	36	PCG.R	DMN
	29	INS.L	SN
	46	CUN.R	VN
	44	CAL.R	VN
	79	HES.L	AN
	83	TPOsup.L	CTC
	76	PAL.R	Subcortical
	33	DCG.L	CTC
Out degree	39	PHG.L	DMN
	84	TPOsup.R	CTC
	77	THA.L	Subcortical
	88	TPOMid.R	DMN
	38	HIP.R	Uncertain
	40	PHG.R	Uncertain
	76	PAL.R	Subcortical
	78	THA.R	Subcortical
	37	HIP.L	Uncertain
	73	PUT.L	Subcortical
	41	AMYG.L	Subcortical
	74	PUT.R	Subcortical
	22	OLF.R	Uncertain
	16	ORBinf.R	DMN
	32	ACG.R	DMN
	75	PAL.L	Subcortical
	28	REC.R	DMN

CTC: cingulo-opercular task control.

Furthermore, the in-depth analysis of consensus connection and hub nodes of functional brain networks highlighted the dysfunction of cortico-subcortical network play a critical role in the pathological mechanisms of early ASD.

CRediT statement

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Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

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Conflict of Interest statement. None declared.

Consent for publication

Not applicable

Data availability

The datasets and custom code utilized and/or generated during the current study can be obtained from the corresponding author upon reasonable request.

References

Alzheimer's Association. 2022 Alzheimer's disease facts and figures. *Alzheimers Dement*. 2022;18(4):700–789. <https://doi.org/10.1002/alz.12638>.

Bamicha V, Drigas. ToM & ASD: the interconnection of theory of mind with the social-emotional, cognitive development of children with autism spectrum disorder. The use of ICTs as an alternative form of intervention in ASD. *Tech Soc Sci J*. 2022;33:42–72.

Barlattani T et al. Autism spectrum disorders and psychiatric comorbidities: a narrative review. *J Psychopathol*. 2023;29:3–24.

Chen S, Zhao Q. Attention-based autism spectrum disorder screening with privileged modality. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2019. p. 1181–1190.

Delong ER, Delong DM, Clarkepearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44(3):837–845.

Ding Y, Xu X, Peng L, Zhang L, Li W, Cao W, Gao X. Wavelet transform-based frequency self-adaptive model for functional brain network. *Cereb Cortex*. 2023;33(22):bhad357–bhad11194.

Easson AK, Fatima Z, McIntosh AR. Functional connectivity-based subtypes of individuals with and without autism spectrum disorder. *Netw Neurosci*. 2019;3(2):344–362.

Friston KJ, Williams S, Howard R, Frackowiak RSJ, Turner R. Movement-related effects in fMRI time-series. *Magn Reson Med*. 1996;35(3):346–355.

Goebel R, Roebroeck A, Kim DS, Formisano E. Investigating directed cortical interactions in time-resolved fMRI data using vector autoregressive modeling and Granger causality mapping. *Magn Reson Imaging*. 2003;21(10):1251–1261.

Guo X, Zhang X, Chen H, Zhai G, Cao Y, Zhang T, Gao L. Exploring the heterogeneity of brain structure in autism spectrum disorder based on individual structural covariance network. *Cereb Cortex*. 2023;33(12):7311–7321.

Jiang X, Yan J, Zhao Y, Jiang M, Chen Y, Zhou J, Xiao Z, Wang Z, Zhang R, Becker B, et al. Characterizing functional brain networks via spatio-temporal attention 4D convolutional neural networks (STA-4DCNNs). *Neural Netw*. 2023;158:99–110.

Langer N, Pedroni A, Gianotti LRR, Hänggi J, Knoch D, Jäncke L. Functional brain network efficiency predicts intelligence. *Hum Brain Mapp*. 2012;33(6):1393–1406.

Lee H, Lee DS, Kang H, Kim BN, Chung MK. Sparse brain network recovery under compressed sensing. *IEEE Trans Med Imaging*. 2011;30(5):1154–1165.

Li Q, Li Y, Liu B, Chen Q, Xing X, Xu G, Yang W. Prevalence of autism spectrum disorder among children and adolescents in the United States from 2019 to 2020. *JAMA Pediatr*. 2022;176(9):943–945.

Li W, Geng C, Chen S. Leave zero out: towards a no-cross-validation approach for model selection arXiv preprint arXiv:2012.13309. 2020.

Li W, Tang Y, Peng L, Wang Z, Hu S, Gao X. The reconfiguration pattern of individual brain metabolic connectome for Parkinson's disease identification. *MedComm*. 2023;4(4):e305.

Li W, Wang Z, Zhang L, Qiao L, Shen D. Remodeling Pearson's correlation for functional brain network estimation and autism spectrum disorder identification. *Front Neuroinform*. 2017;11:55.

Li W, Xu X, Wang Z, Peng L, Wang P, Gao X. Multiple connection pattern combination from single-mode data for mild cognitive impairment identification. *Front Cell Develop Biol*. 2021;9:782727.

Li W et al. Towards a better estimation of functional brain network for mild cognitive impairment identification: a transfer learning view. *IEEE J Biomed Health Inform*. 2019;24(4):1160–1168.

Liao W, Wu GR, Xu Q, Ji GJ, Zhang Z, Zang YF, Lu G. DynamicBC: a MATLAB toolbox for dynamic brain connectome analysis. *Brain Connect*. 2014;4(10):780–790.

Liu J et al. Replicable patterns of memory impairments in children with autism and their links to hyperconnected brain circuits. *Biol Psychiatry: Cogn Neurosci Neuroimaging*. 2023;8(11):1113–1123.

Ma K, Huang S, Wan P, Zhang D. Optimal transport based pyramid graph kernel for autism Spectrum disorder diagnosis. *Pattern Recogn*. 2023;109716:109716.

McLoughlin G. Genetic overlap between midfrontal theta signals and attention-deficit/hyperactivity disorder and autism spectrum disorder in a longitudinal twin cohort. *Biol Psychiatry*. 2023.

Ng B et al. *A novel sparse graphical approach for multimodal brain connectivity inference*. Berlin Heidelberg: Springer; 2012.

- Peng L, Liu X, Ma D, Chen X, Xu X, Gao X. The altered pattern of the functional connectome related to pathological biomarkers in individuals for autism spectrum disorder identification. *Front Neurosci*. 2022;16:913377.
- Qiang N, Gao J, Dong Q, Li J, Zhang S, Liang H, Sun Y, Ge B, Liu Z, Wu Z, et al. A deep learning method for autism spectrum disorder identification based on interactions of hierarchical brain networks. *Behav Brain Res*. 2023;452:114603.
- Roebroeck A, Formisano E, Goebel R. Mapping directed influence over the brain using Granger causality and fMRI. *NeuroImage*. 2005;25(1):230–242.
- Sato W, Uono S. The atypical social brain network in autism: advances in structural and functional MRI studies. *Curr Opin Neurol*. 2019;32(4):617–621.
- Shao L, Fu C, Chen X. A heterogeneous graph convolutional attention network method for classification of autism spectrum disorder. *BMC Bioinform*. 2023;24(1):1–17.
- Shi Z, Jiang B, Liu T, Wang L, Pei G, Suo D, Zhang J, Funahashi S, Wu J, Yan T. Individual-level functional connectomes predict the motor symptoms of Parkinson's disease. *Cereb Cortex*. 2023;33(10):6282–6290.
- Stevens E, Dixon DR, Novack MN, Granpeesheh D, Smith T, Linstead E. Identification and analysis of behavioral phenotypes in autism spectrum disorder via unsupervised machine learning. *Int J Med Inform*. 2019;129:29–36.
- Unruh KE, Bartolotti JV, McKinney WS, Schmitt LM, Sweeney JA, Mosconi MW. Functional connectivity of cortical-cerebellar networks in relation to sensorimotor behavior and clinical features in autism spectrum disorder. *Cereb Cortex*. 2023;33(14):bhad177–bhad9002.
- Wang J, Wang Q, Peng J, Nie D, Zhao F, Kim M, Zhang H, Wee CY, Wang S, Shen D. Multi-task diagnosis for autism spectrum disorders using multi-modality features: a multi-center study. *Hum Brain Mapp*. 2017;38(6):3081–3097.
- Wee CY, Yap PT, Shen D. Diagnosis of autism spectrum disorders using temporally distinct resting-state functional connectivity networks. *CNS Neurosci Ther*. 2016;22(3):212–219.
- Wen G, Cao P, Bao H, Yang W, Zheng T, Zaiane O. MVS-GCN: a prior brain structure learning-guided multi-view graph convolution network for autism spectrum disorder diagnosis. *Comput Biol Med*. 2022;142:105239.
- Xu X, Li W, Tao M, Xie Z, Gao X, Yue L, Wang P. Effective and accurate diagnosis of subjective cognitive decline based on functional connection and graph theory view. *Front Neurosci*. 2020;14:577887.
- Xu X, Wang T, Li W, Li H, Xu B, Zhang M, Yue L, Wang P, Xiao S. Morphological, structural, and functional networks highlight the role of the cortical-subcortical circuit in individuals with subjective cognitive decline. *Front Aging Neurosci*. 2021;13:688113.
- Yang C, Gao X, Liu N, Sun H, Gong Q, Yao L, Lui S. Convergent and distinct neural structural and functional patterns of mild cognitive impairment: a multimodal meta-analysis. *Cereb Cortex*. 2023;33(14):bhad167–bhad8889.
- Yu R, Qiao L, Chen M, Lee SW, Fei X, Shen D. Weighted graph regularized sparse brain network construction for MCI identification. *Pattern Recogn*. 2019;90:220–231.
- Zhang L, Pang M, Liu X, Hao X, Wang M, Xie C, Zhang Z, Yuan Y, Zhang D. Incorporating multi-stage diagnosis status to mine associations between genetic risk variants and the multi-modality phenotype network in major depressive disorder. *Front Psychiatry*. 2023;14:1139451.
- Zhang L, Wang M, Liu M, Zhang D. A survey on deep learning for neuroimaging-based brain disorder analysis. *Front Neurosci*. 2020;14:779.
- Zheng Q, Zhu J, Li Z, Tian Z, Li C. Comprehensive multi-view representation learning. *Inf Fusion*. 2023;89:198–209.
- Zhou Z, Chen X, Zhang Y, Qiao L, Yu R, Pan G, Zhang H, Shen D. Brain Network Construction and Classification Toolbox (Brain-NetClass) arXiv preprint arXiv:1906.09908. 2019.